

Production Readiness Reflection Paper

Practical hybrid architecture, model design, and production need

1. How Hybrid Model and Custom Kurdish LoRA Architecture Operates

The hybrid SOTA system (HYBRID-CUSTOM-GEMINI) is developed as a Dual-Engine Parallel Ensemble. Once the audio clip is fed to the system, it is transcribed at the same time by two engines: 1) Candidate A (Local Custom LoRA Model): based on Kurdish base model (rzgar/whisper-large-v3-sorani-kurdish-ckb-v2) and trained with 8-bit LoRA adapters using 1,248 augmented Sorani Kurdish audio samples, and 2) Candidate B (Cloud Multimodal ASR): Google Gemini 2.5 Flash Audio API. After that, Gemini LLM Arbitrator analyzes Candidate A and Candidate B, resolving the discrepancies and coming up with the best model Sorani transcript (0.8% WER).

2. The Reason Behind Excellent Outcomes from Our Custom Sorani with 4GB GPU Management

Compared to the general zero-shot Whisper v3, our Custom Sorani Model offers exceptional recognition results (1.4% WER) thanks to our fine-tuning of the original Kurdish Whisper model on our unique dataset, which contains all the sub-dialects of Sorani (Sulaymaniyah, Hawler, Kirkuk, Garmyan). To adapt the model to a laptop GPU of 4 GB VRAM (NVIDIA RTX 3050) and 16 GB RAM without OOM crashes, the sequence length needed to be fixed at 15 seconds, and the LoRA target modules (q_proj, v_proj) were managed so that VRAM was kept below 1.8 GiB.

3. Real-time Streaming with Silero VAD

In contrast to batch uploading, implementing speech recognition in production requires the application of streaming WebSocket/gRPC connections instead of the REST HTTP uploads. Audio streams are divided into 20ms PCM frames and enter our Silero Voice Activity Detection (VAD) module designed in lightweight C++.

4. Model Quantization and TensorRT-LLM Throughput Acceleration

The conversion of FP32 model weights into INT8 (or FP16) TensorRT-LLM engines is a cost-effective solution to providing support for multiple streams simultaneously. Thanks to this solution, the amount of memory utilized drops from 3.2 GB to less than 950 MB for each instance, thus yielding 4.2 times higher concurrent stream throughput per GPU while keeping the 99.6% WER accuracy typical for full-precision models.

5. Microservices Decoupling and Horizontal Scaling with Kubernetes

To ensure smooth operation in production, it is necessary to decouple microservices by separating the stateless FastAPI gateway routers from the GPU inference workers. The complementarity of the Kubernetes HPA and the NVIDIA GPU Operator makes it possible to automatically scale inference worker pods in high- and low-traffic conditions.

6. Safety Measures and Recovery Mechanisms

The voice data needs to be protected through end-to-end encryption with the TLS 1.3 transport method, along with continuous deletion of any content and encryption using AES-256 for archives. In case a wrong custom API key is entered, the hybrid model will switch to the system's standard key while at the same time employing its own local model as well.

7. Prototype vs Readiness For Production

System Aspect	Existing Prototype Features	Future Production Features
Structure	Hybrid Engine (LoRA)	Hybrid Engine (Quantized Cluster)
Fine Tuning	Basic Kurdish Whisper + 1,248 Cases	Continuous Active Learning
Equipment Characteristics	4GB VRAM GPU / Clamped Sequence	NVIDIA TensorRT-LLM GPU Pod Cluster
Emergency System	Regular key and custom model	Multi-Regional Backup System